

Vehicle Cut-In Detection and Platoon Reconfiguration

Description:

A platoon of autonomous trucks is driving along a highway, keeping a pre-defined inter-vehicle distance. Suddenly a non-platoon vehicle cuts in the gap between two platoon trucks.

If the platoon detects an intruding vehicle, the platoon shall notify other platoon members, increase following distances to maintain safety, and suspend normal platoon operation for a time. The trucks re-form the platoon and operate normally after the intruding vehicle leaves the gap in the platoon.

Assumptions:

- All trucks are equipped with radar and camera sensors.
- Vehicle-to-vehicle communication is available.
- Road markings are visible and lane detection is functioning.
- Weather conditions allow reliable sensing.
- The platoon leader remains operational throughout the scenario.

Functional Requirements:

- The system shall continuously monitor the platoon gap.
- The system shall detect a vehicle entering the platoon gap.
- The detecting truck shall notify all platoon members.
- The platoon shall increase inter-vehicle spacing after cut-in detection.
- The platoon shall enter reconfiguration mode.
- The platoon leader shall coordinate platoon reformation.
- The platoon shall return to normal driving mode after reformation.

Non-functional Requirements:

- Detection accuracy shall be greater than 95%.
- Communication reliability shall exceed 99%.
- The system shall operate continuously at highway speeds.
- The reconfiguration process shall minimize traffic disruption.

Timing Requirements:

- Cut-in detection shall occur within 200 ms
- Notification messages shall be transmitted within 100 ms.
- Distance adjustment shall begin within 500 ms.
- Platoon reformation shall complete within 10 seconds after the vehicle leaves.

Safety Requirements:

- The platoon shall maintain a safe following distance at all times.
- No collision shall occur during platoon reconfiguration.
- Communication failure shall trigger a safe fallback operation.
- The platoon shall not reform while the intruding vehicle remains in the gap.